

FIITJEE

**ALL INDIA TEST SERIES**

**CONCEPT RECAPITULATION TEST – I**

**JEE (Main)-2024**

**TEST DATE: 20-01-2024**

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**ANSWERS, HINTS & SOLUTIONS**

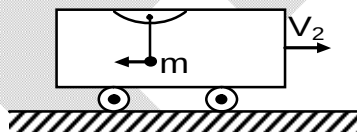
**Physics**

PART – A

SECTION – A

1. B  
Sol. Spring forces does not change instantaneously thus for  $m_1$  ;  $a_1 = a_0$   
For  $m_2$   $F_{sp} = m_2 a_2$  .....(i) instantaneously after  $F_2$  is withdrawn  
Initially  $F_{sp} - F_2 - m_2 a_0 = 0$   $F_{sp} \rightarrow \square$   
 $F_{sp} = F_2 + m_2 a_0$  – (ii)  
From (i) and (ii)  $a_2 = \frac{F_2}{m_2} + a_0$

2. A  
Sol.  $WD = \Delta K$



$$mgl = \frac{1}{2} \frac{m \cdot 6m}{m + 6m} (V_1 + V_2)^2 - 0$$

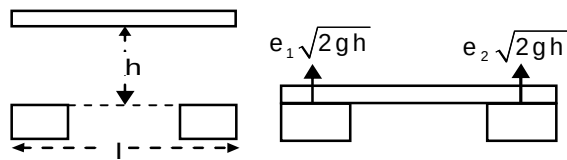
$$\therefore V_1 + V_2 = \sqrt{\frac{7}{3}gl}$$

3. A

Sol.  $\tan 30^\circ = \frac{eV \sin 60^\circ}{\tan 60^\circ}$

$$e = \frac{\tan 30^\circ}{\tan 60^\circ} = \tan^2 30^\circ$$

4. B

Sol. 

$$V_{cm} = \frac{e_1 + e_2}{2} \sqrt{2gh}$$

$$h_{max} = \frac{(V_{cm})^2}{2g} = \frac{(e_1 + e_2)^2 h}{4}$$

5. B

Sol.  $\therefore \sin^3 \omega t = \frac{1}{4} (3 \sin \omega t - \sin 3\omega t)$

6. B

Sol. Initial length (circumference) of the ring =  $2\pi r$   
 Final length (circumference) of the ring =  $2\pi R$   
 Change in length =  $2\pi R - 2\pi r$

$$\text{Strain} = \frac{\text{Change in length}}{\text{Original length}} = \frac{2\pi(R-r)}{2\pi r} = \frac{R-r}{r}$$

Now Young's modulus,  $E = \frac{F/A}{l/L} = \frac{F/A}{(R-r)/r}$

$$\therefore F = AE \left( \frac{R-r}{r} \right)$$

7. A

Sol.  $V_0 = V_{ball} = \sqrt{2gR}$

$$\int T \cdot dt = -\frac{m}{2} (V - V_0) \quad \dots(i)$$

$$\int T \cdot dt \cdot R = \frac{1}{2} M \cdot R^2 \cdot \frac{V}{R} \quad \dots(ii)$$

$$-\frac{M}{2} (V - V_0) = \frac{M}{2} V$$

$$2V = V_0 \Rightarrow V = \frac{V_0}{2} = \frac{\sqrt{2gR}}{2} = \sqrt{\frac{gR}{2}}$$

$$\omega = \frac{V}{R} = \sqrt{\frac{g}{2R}}$$

8. D

Sol. In the ideal case that we normally consider each collision transfers twice the magnitude of its normal momentum. On the face EFGH, it transfers only half of that.

9. A

Sol. Slope of  $V - T$  graph is  $\frac{dV}{dT} = \frac{nR}{P}$

10. C

Sol. Electric field is directed in the direction of fall of potential  $\left[ E = -\frac{dV}{dr} \right]$

11. C

Sol.  $E = \rho \frac{i}{A}$   
 $E = \frac{kr^2}{R} \frac{i}{\pi r^2}$   
 $E = \frac{ki}{\pi R}$

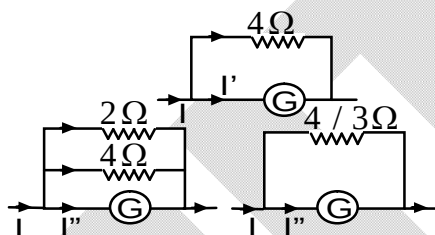
12. D

Sol.  $I' = \frac{4}{4+G} I \Rightarrow \frac{4}{4+G} = \frac{1}{5}$  (given) where  $G = 16\Omega$

$$I'' = \frac{4/3}{\frac{4}{3} + 16} I$$

$$I'' = \frac{1}{13} I$$

$$\Rightarrow I'' = \frac{0.65}{13} = 0.05$$



13. C

Sol. Work done to form a bubble of radius  $R$

$$W_1 = 8\pi R^2 T_1$$

Work done to form a bubble of radius  $2R$

$$W_2 = 8\pi (2R)^2 T_2 = 32\pi R^2 T_2 \quad \therefore \frac{W_1}{W_2} = \frac{T_1}{4T_2}$$

If surface tension of soap solution is same, then  $W_2 = 4W_1$

But in the problem the temperature of solution is increased. So its surface tension decreases.

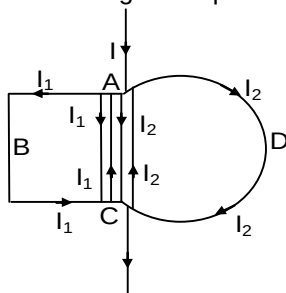
$$\therefore W_2 < 4W_1$$

14. C

Sol. Least count in B is highest of all the measurements hence accuracy is least.

15.

B

 Sol. Introducing two equal and opposite current  $I_1$  and also  $I_2$  between A & C.


Force on ABCA closed loop zero

Force on ADCA closed loop zero

 Force on Extra  $I_1$  &  $I_2$ 

$$F = (I_1 + I_2) lB = I lB$$

16.

B

 Sol. Phase difference for current in inductor and capacitor is  $180^\circ$ 

$$i = i_L \sim i_C$$

17.

B

 Sol. In continuous spectrum of X – ray the minimum wave length is given by  $\lambda_{\min} = \frac{hc}{eV}$  and the

$$\text{characteristic X – ray } \lambda \propto \frac{1}{(Z-1)^2}$$

18.

A

 Sol.  $\lambda$  longest  $\Rightarrow E_{\min}$ 

 For Balmer  $\lambda$  longest is for  $n = 2 \rightarrow 3$ 
 $\Rightarrow \lambda$  longest Balmer  $>$   $\lambda$  longest Lyman

19.

B

Sol. For positive half cycle the junction diode is forward biased.

20.

C

Sol. Flux coming out of the cube

$$\phi_1 = \frac{\lambda \cdot a \sqrt{3}}{\epsilon_0} \quad \dots(i)$$

and from sphere

$$\phi_2 = \frac{\lambda \cdot 2a}{\epsilon_0} \quad \dots(ii)$$

### SECTION – B

21.

274

 Sol.  $VS = 4$ ,  $MS = 2.7$ 

$$= MS + VS \times LC$$

$$= 2.7 + 0.04 = 2.74$$

22. 25

Sol. Suppose in equilibrium wire PQ lies at a distance  $r$  above the wire AB

$$\text{Hence in equilibrium } mg = Bil \Rightarrow mg = \frac{\mu_0}{4\pi} \left( \frac{2i}{r} \right) \times il$$

$$\Rightarrow 10^{-3} \times 10 = 10^{-7} \times \frac{2 \times (50)^2}{r} = 0.5 \Rightarrow r = 25 \text{ mm}$$

23. 30

Sol. The area of the  $a - t$  graph gives the change in the velocity of the body. The body will acquire the maximum speed till this acceleration that is the maximum velocity will be at  $t = 8 \text{ sec}$ 

$$\therefore v_{\max} = \left( \frac{8+4}{2} \right) \times 5 = 30$$

24. 75

Sol. For normal incidence on second surface  $r_2 = 0$ 

$$\therefore r_1 = A \text{ and}$$

$$\therefore \mu = \frac{\sin i}{\sin r}$$

$$\text{For small angle } \mu = \frac{i}{r} = \frac{\theta}{A} \Rightarrow 1.5 = \frac{\theta}{5r!} \frac{n!}{(n-r)!}$$

25. 14

Sol.  $\phi = 2\text{eV}$ 

$$\frac{hc}{\lambda_1} = 8\text{eV} \quad T_2 = 2T_1$$

If  $\lambda_1$  is the wavelength corresponding to maximum intensity at  $T_1$  &  $T_2$  at  $T_2$ ; Then  $\lambda_2 = \lambda_1 / 2$   
(by Wein's displacement Law)

$$\frac{hc}{\lambda_2} = \frac{2hc}{\lambda_1} = 16 \text{ eV}$$

$$\phi = 2 \text{ eV} \quad \therefore K.E._{\max} = \frac{hc}{\lambda} - \phi = 14 \text{ eV}$$

26. 10

Sol.  $Q - \text{Value} = \frac{1}{R} \sqrt{\frac{L}{C}}$ 

27. 13

Sol.  $i_L = \frac{50}{10 \times 10^3} = 5\text{mA}$ 

$$i = \frac{140 - 50}{5 \times 10^3} = 18\text{mA}$$

$$\therefore i_z = i - i_L = 13\text{mA}$$

28. 1200

Sol.  $RT = 200V + 1$ 

$$\Delta U = \left( \frac{R}{\gamma - 1} \right) (T_f - T_i) = 1 \frac{R}{\frac{4}{3} - 1} (T_f - T_i)$$

$$= 3R(T_f - T_i) = 3 \times 200(V_f - V_i)$$

$$= 1200 \text{ J}$$

29. 1

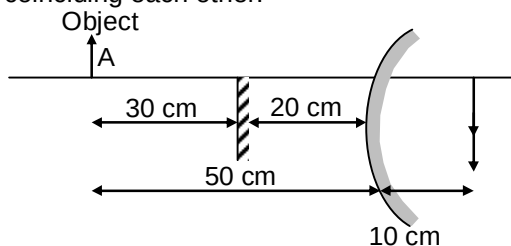
Sol. Thrust =  $\rho a.v^2$

$$= \rho a.2gh = 10^3 \times 10^{-3} \times 2 \times 10 \times 50 \times 10^{-3}$$

$$= 1 \text{ N}$$

30. 25

Sol. Since there is no parallax, it means that both images (By plane mirror and convex mirror) coinciding each other.



According to property of plane mirror it will form image at a distance of 30 cm behind it. Hence for convex mirror  $u = -50 \text{ cm}$ ,  $v = +10 \text{ cm}$

By using  $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{+10} + \frac{1}{-50} = \frac{4}{50}$

$$\Rightarrow f = \frac{25}{2} \text{ cm} \Rightarrow R = 2f = 25 \text{ cm}$$

# Chemistry

## PART – B

### SECTION – A

31. C

Sol.  $\lambda = \frac{h}{\sqrt{2mqV}}$ ;  $q = 2 \times 1.06 \times 10^{-19} \text{ C}$

$V = V \text{ volts}$

$m = 4 \text{ amu} = 4 \times 1.67 \times 10^{-27} \text{ kg}$

$$\left[ \lambda = \frac{0.101}{\sqrt{V}} \text{ \AA} \right]$$

32. D

Sol. Let aqueous V. P. be  $p \text{ mm of Hg}$

Initial conditions

$P_1 \text{ (dry gas)} = (750 - p) \text{ mm}$

$V_1 = 100 \text{ mL}$

$T_1 = 15 + 273 = 288 \text{ K}$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}, \frac{(750 - p)}{288} \times 100 = \frac{760 \times 91.9}{273}$$

$(750 - p) = 736.8$

$P = 13.2 \text{ mm}$

NTP conditions

$P_2 = 760 \text{ mm}$

$V_2 = 91.9 \text{ mL}$

$T_2 = 273 \text{ K}$

33. D

Sol.  $K_p = K_c (RT)^{\Delta n}$ ;  $K_c = \frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$

34. A

Sol.  $\text{CN}^-$  is strong field ligand Greater the number of  $\text{CN}^-$ , more will be the strength of ligands more will be the value of  $\Delta_o$  (crystal field splitting energy).

35. D

Sol. Decomposition of  $\text{N}_2\text{O}$  on hot platinum is zero order reaction so unit rate constant =  $\text{mol L}^{-1} \text{Time}^{-1}$ .

Acidic hydrolysis of ester is first order reaction, so unit of rate constant =  $\text{Time}^{-1}$ .

Alkaline hydrolysis of ester is second order reaction, so unit of rate constant =  $\text{mol}^{-1} \text{L Time}^{-1}$ .

Reduction of  $\text{FeCl}_3$  by  $\text{SnCl}_2$ , is third order reaction, so unit of rate constant =  $\text{mol}^{-2} \text{L}^2 \text{Time}^{-1}$ .

36. D

Sol. As in  $\text{X}_3(\text{YZ}_4)_2$  the sum of all oxidation numbers is zero  
 $+6 + 10 - 16 = 0$

37. C

Sol. The jump in I.P values is noticed during successive removal of electron when valence shell changes. As in 4 and 5 IP Values the difference is maximum so it is carbon.

38. A

Sol. (A) Increasing order of magnetic moment will be  $d^1, d^3, d^4, d^5$ . So this one is not correct  
 (B) as the oxidation state of metal increases acidic strength increases so this one is correctly matched  
 (C) Mn have highest oxidation state of +7

39. C

Sol.  $2K_2(Hgl_4) + NH_4OH + 3NaOH \longrightarrow I-Hg-O-Hg-NH_2 + 4KI + 3NaI + 3H_2O$   
 Nesseler's reagent

40. B

Sol. The number of molecules with energy greater than the activation energy at  $T_2$  is given by  $Q + R$ .

Total number of molecules is  $P + Q$ . The required fraction is  $\frac{Q+R}{P+Q}$ .

41. C

Sol. Amidines (I) are most basic due to equal resonating structure after protonation, amides are least basic as lone pair on N atom is in conjugation with carbonyl group. Secondary amines are more basic than primary amines.

42. C

Sol.  $R_f = \frac{\text{Distance travelled by substance}}{\text{Distance travelled by eluent}}$

**For A:**

$$0.75 = \frac{\text{Distance travelled by substance A}}{8}$$

Distance travelled by substance A = 6 cm

**For B:**

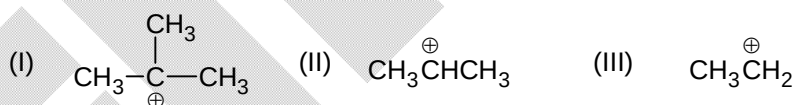
$$0.50 = \frac{\text{Distance travelled by substance B}}{8}$$

Distance travelled by substance B = 4 cm

Difference in the height of A and B =  $6 - 4 = 2$  cm

43. A

Sol. Due to more stable carbocation

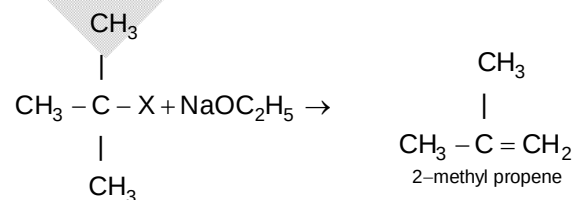


44. D

Sol. 1 and 3 are same.

45. B

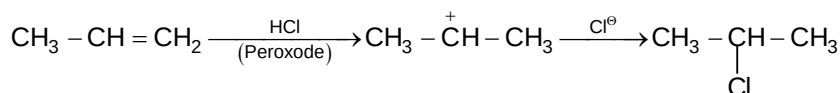
Sol. The tertiary alkyl halide undergoes elimination reaction to give alkenes





46. B

Sol. HCl is not affected by peroxide so major product formed by E.A.R.

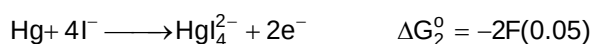
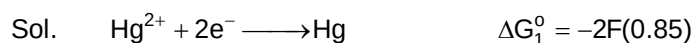


47. D

Sol.  $\text{pI} = \frac{\text{pK}_a \text{ of acidic group} + \text{pK}_a \text{ of basic group}}{2}$

$$= \frac{y + (14 - x)}{2}$$

48. D



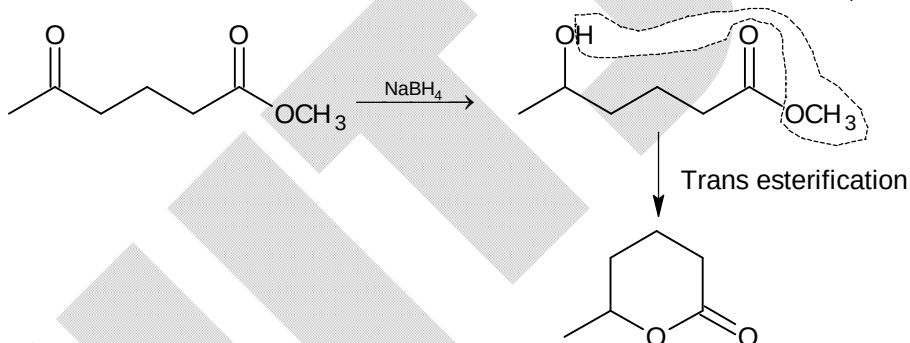
so



$$0.90 = \frac{0.06}{2} \log K_f$$

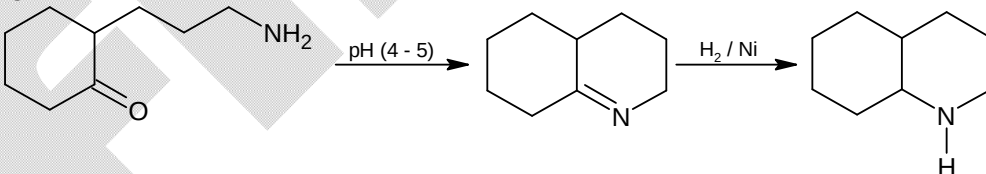
$$K_f = 1.0 \times 10^{30}$$

49. C

Sol. Reduction of ester group is not affected by weak reducing agent  $\text{NaBH}_4$ 

50. C

Sol.



## SECTION - B

51. 2

Sol. mole of  $\text{HCl} = \frac{1}{100} \times 10^{-3} = 10^{-5}$  mole

conc of  $\text{HCl} = 10^{-5} \text{ M}$

so  $\text{pH} = 5$

change in  $\text{pH} = 7 - 5 = 2$

52. 328

Sol. Applying Hess's Law

$$\Delta_f H^\circ = \Delta_{\text{sub}} H + \frac{1}{2} \Delta_{\text{diss}} H + I.E. + E.A + \Delta_{\text{lattice}} H$$

$$-617 = 161 + 520 + 77 + E.(A) + (-1047)$$

$$E.(A) = -167 + 289 = -328 \text{ kJ mol}^{-1}$$

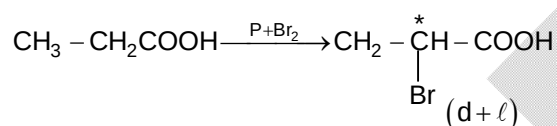
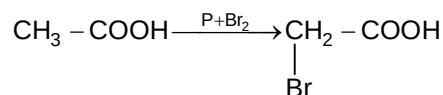
 $\therefore$  electron affinity of fluorine

$$= -328 \text{ kJ mol}^{-1}$$

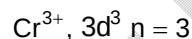
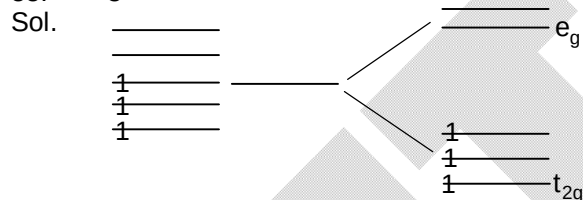
53. 9

$$\text{Sol. } \text{pI} = \left( \frac{\text{pK}_{a_2} + \text{pK}_{a_3}}{2} \right) = \frac{8.0 + 10.0}{2} = 9.0$$

54. 3

 Sol. Formic acid and benzoic acid do not contain  $\alpha$  - H atoms so they cannot give HVZ reaction


55. 3



56. 12

Sol. II, III, IV, V, VII, VIII, IX, X, XI, XIII, XIV, XVI can show geometrical isomerism.

57. 550

$$\text{Sol. } \lambda^\circ(\text{Ba}(\text{OH})_2) = \lambda^\circ(\text{BaCl}_2) + 2\lambda^\circ(\text{NaOH}) - 2\lambda^\circ(\text{NaCl})$$

$$\lambda^\circ(\text{Ba}(\text{OH})_2) = 300 + 2(250) - 2(125)$$

$$\lambda^\circ(\text{Ba}(\text{OH})_2) = 300 + 500 - 250$$

$$\lambda^\circ(\text{Ba}(\text{OH})_2) = 550 \text{ mho cm}^2 \text{ mol}^{-1}$$

58. 12

$$\text{Sol. } V_1 = \frac{2 \times 0.0821 \times 300}{8.21} = 6.0 \text{ L}$$

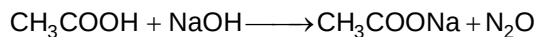
$$V_2 = \frac{2 \times 0.0821 \times 300}{2.73} = 18.0 \text{ L}$$

$$W_{\text{irr, isothermal}} = -P_{\text{ext}} \times \Delta V = -1 \times (18 - 6) \\ = -12 \text{ lt-atm}$$

59. 3

Sol. Milli. moles of NaOH =  $30 \times 0.2 = 6 \text{ m.m}$ m. m of  $\text{CH}_3\text{COOH} = 50 \times 0.2 = 10 \text{ m.m}$ 

Now:



$$\begin{array}{ccc} 10 & 6 & 0 \\ 10 - 6 = 0 & 6 - 6 = 0 & 6 \end{array}$$

$$\text{pH} = -\log(2 \times 10^{-5}) + \log \frac{6}{4} \\ = 4.87$$

Suppose 'v' mL of NaOH is added then

m. m of  $\text{CH}_3\text{COONa} = (6 + v \times 0.2)$ m. m of  $\text{CH}_3\text{COOH} = (4 - v \times 0.2)$ 

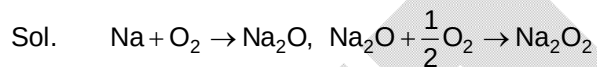
$$5 = -\log(2 \times 10^{-5}) + \log \left( \frac{6 + 0.2v}{4 - 0.2v} \right)$$

$$0.3010 = \log \left( \frac{6 + 0.2v}{4 - 0.2v} \right)$$

$$\text{So } \left( \frac{6 + 0.2v}{4 - 0.2v} \right) = 2 \Rightarrow v = 3.33 \text{ mL} = \frac{10}{3} \text{ mL}$$

So x = 3

60. 1

It exists as  $2\text{Na}^+$  and  $\text{O}_2^{2-}$  So oxidation state of O is -1.

# Mathematics

## PART – C

### SECTION – A

61. C

$$\text{Sol. } y = \tan^{-1}\left(\frac{5x-x}{1+5x(x)}\right) + \tan^{-1}\left(\frac{x+\frac{2}{3}}{1-\frac{2x}{3}}\right) = (\tan^{-1}5x - \tan^{-1}x) + \left(\tan^{-1}x + \tan^{-1}\frac{2}{3}\right)$$

$$y' = \frac{5}{1+25x^2}$$

$$\therefore a = 5$$

62. A

$$\begin{aligned} \text{Sol. } \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{{}^nC_k}{n^k} \int_0^1 \left[ \lim_{n \rightarrow \infty} {}^nC_k \left(\frac{x}{n}\right)^k x^2 \right] dx &= \left[ \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n x^2 \right] dx \\ &= \int_0^1 e^x x^2 dx = \int_0^1 e^x (x^2 + 2x) dx - 2 \int_0^1 e^x (x+1) dx + 2 \int_0^1 e^x dx \\ &= e^x (x^2 - 2x + 2) \Big|_0^1 = e - 2 \end{aligned}$$

63. B

- Sol. (A) False because if  $g(x)$  is  $\sin(2\pi x)$ .  
 (B) Take logarithm on both sides and differentiate once to get the expression.  
 (C) Obviously, false.  
 (D) Statement is correct for  $|f(x)|$  but not for  $f(x)$

64. A

$$\text{Sol. Using limit of substitution, Put } \frac{x^n}{e^x} = t.$$

$$\text{Now, as } x \rightarrow \infty, \frac{x^n}{e^x} \rightarrow 0$$

$$\text{So, } \lim_{t \rightarrow 0} \frac{2^t - 3^t}{t}$$

65. B

$$\begin{aligned} \text{Sol. } (f(x)-1)^2 (f(x)-x^3) &= 0 \\ \therefore f(x) &= x^3 \\ f'(x) &= 3x^2 \end{aligned}$$

66. C

$$\text{Sol. } I = \int \frac{3(\tan x - 1)\sec^2 x}{(\tan x + 1)\sqrt{\tan^3 x + \tan^2 x + \tan x}} dx = 3 \int \frac{(t-1)}{(t+1)\sqrt{t^3+t^2+t}} dt$$

$$= 3 \int \frac{\left(1 - \frac{1}{t^2}\right)}{\left(t + \frac{1}{t} + 2\right) \sqrt{t + \frac{1}{t} + 1}} dt \quad \text{Let } t + \frac{1}{t} + 1 = z^2 \Rightarrow \left(1 - \frac{1}{t^2}\right) dt = 2z dz$$

67. D

Sol.  $\int \left(\frac{y+1}{y}\right) dy = \int e^x (\sin 2x - \cos^2 x) dx$   
 $\Rightarrow y + \ln y = -e^x \cos^2 x + C$

68. A

Sol.  $y = f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 2x^2 & \text{if } x \geq 0 \end{cases}$   
 $\therefore f(x)$  is continuous and derivable.

69. B

Sol.  $\cos 2\theta = \frac{1}{3}$  (Given)  
 $\therefore \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{3} \Rightarrow 3 - 3\tan^2 \theta = 1 + \tan^2 \theta$   
 $\Rightarrow 4\tan^2 \theta = 2$  or  $\tan^2 \theta = \frac{1}{2} \Rightarrow \tan^8 \theta = \frac{1}{16}$   
 $\Rightarrow 2\cos 2\alpha - 3\cos \alpha = 32 \cdot \left(\frac{1}{16}\right) = 2$

70. B

Sol. The difference of radius is equal to diameter of circle.  
 Radius of given circle  $\Rightarrow (x+1)^2 + (y+2)^2 - 9$ , is 3  
 $\therefore$  Difference = 6

71. C

Sol.  $A = (2\alpha + 1, 3\alpha + 2, 4\alpha + 3)$   
 $B = (\beta + 2, 2\beta - 1, 3\beta - 2)$   
 $\therefore \frac{2\alpha - \beta - 1}{\frac{2}{(1)}} = \frac{3\alpha - 2\beta + 3}{\frac{1}{(2)}} = \frac{4\alpha - 3\beta + 5}{\frac{1}{(3)}}$   
 Solving (1) and (2)  $\Rightarrow 4\alpha - 3\beta + 7 = 0$   
 Solving (2) and (3)  $\Rightarrow \alpha - \beta + 2 = 0$   
 $\alpha = -1, \beta = 1, A(-1, -1, -1); B(3, 1, 1)$   
 $AB = 2\sqrt{6}$

72. A

Sol.  $2\vec{a} - 3\vec{b} + 6\vec{c} = \vec{0} \Rightarrow 2\vec{a} - 3\vec{b} = -6\vec{c} \Rightarrow |2\vec{a} - 3\vec{b}|^2 = 36|\vec{c}|^2$   
 $4a^2 + 9b^2 - 12\vec{a} \cdot \vec{b} = 36c^2$   
 $4a^2 + 9b^2 - 12.ab \cos \theta = 36c^2$

$$16b^2 + 9b^2 - 12 \cdot 2b^2 \cos \theta = 36 \frac{1}{4} b^2$$

$$25 - 24 \cos \theta = 9 \Rightarrow \cos \theta = \frac{2}{3}$$

73. A

Sol. Use Polar coordinates.

74. C

Sol. Use basic concepts.

75. D

Sol. Hence required number of ways  $\frac{{}^{n-6}C_4 \cdot n}{5} = 36$

Which is satisfied by  $n = 12$

76. C

Sol. Let  $(1 + 3x + 2x^2)^6 = \sum_{k=0}^{12} a_k x^k$  ..... (i)

$$(1 + 3x + 2x^2)^6 = (1 + 2x)^6 (1 + x)^6$$

$$\text{Coefficient of } x^{12} = {}^6C_6 2^6 \cdot {}^6C_6 = 2^6 = a_{12}$$

$$\begin{aligned} \text{Coefficient of } x^{11} &= ({}^6C_6 2^6 \cdot {}^6C_5) + ({}^6C_5 2^5 \cdot {}^6C_6) \\ &= (6 \times 2^6) + (6 \times 2^5) = 9 \times 2^6 \end{aligned}$$

Put  $x = 1$  and  $-1$  in equation (i) and adding, we get

$$(1 + 3 + 2)^6 + (1 - 3 + 2)^6 = (a_0 + a_1 + a_2 + \dots + a_{12}) + (a_0 - a_1 + a_2 - a_3 + \dots + a_{12})$$

$$\Rightarrow a_0 + a_2 + \dots + a_{12} = \frac{6^6}{2}$$

$$\Rightarrow \frac{\sum_{k=0}^6 a_{2k}}{a_{12}} = \frac{3^6}{2}$$

77. B

Sol.  $\sum \alpha = 0$ ;  $\sum \alpha\beta = -p$ ;  $\alpha\beta\gamma = -1$

$$|A + B + C| = \begin{vmatrix} \alpha^3 + \beta^3 + \gamma^3 + 4 & 0 & 0 \\ 0 & \alpha^3 + \beta^3 + \gamma^3 + 4 & 0 \\ 0 & 0 & \alpha^3 + \beta^3 + \gamma^3 + 4 \end{vmatrix}$$

$$= (\alpha^3 + \beta^3 + \gamma^3 + 4)^3 = 1$$

$$\alpha + \beta + \gamma = 0$$

$$\therefore \sum \alpha^3 = 3\alpha\beta\gamma = -3$$

78. A

Sol. Given question becomes

$$= \det A^1 \cdot \det \left( \text{adj} \left( \frac{\text{adj } B}{|B|} \right) \right) \cdot \det \left( \text{adj} \left( \frac{\text{adj } A}{|A|} \right) \right)$$

$$= \frac{1}{|A|} |\text{adj}(\text{adj } B)| \cdot |\text{adj adj}(A)|$$

$$= \frac{1}{|A|} \cdot |B|^4 (1) |A|^4 = 8$$

79. A

Sol. Assuming  $\arg z_1 = \theta$  and  $\arg z_2 = \theta + \alpha$ 

$$\frac{az_1}{bz_2} + \frac{bz_2}{az_1} = \frac{a|z_1|e^{i\theta}}{b|z_2|e^{i(\theta+\alpha)}} + \frac{b|z_2|e^{i(\theta+\alpha)}}{a|z_1|e^{i\theta}}$$

$$= e^{i\alpha} + e^{-i\alpha} = 2\cos\alpha$$

80. C

Sol. Use basic concepts.

**SECTION - B**

81. 743

Sol.  $(EM)^T = 20I$ 

Take transpose on both sides

$$EM = 20I$$

$$(E + M)^T = 17(E - M)^T$$

$$E^T + M^T = 17(E^T - M^T)$$

$$16E^T = 18M^T$$

Take transpose on both sides

$$16E = 18M$$

$$\Rightarrow E^2 = \frac{9}{8}EM \text{ \& } M^2 = \frac{8}{9}EM$$

$$\therefore E^2 + M^2 = \left(\frac{9}{8} + \frac{8}{9}\right)EM = \frac{145}{72}20I = \frac{725}{18}I$$

82. 7

$$\text{Sol. } \int_1^4 x(x - f^{-1}(x))dx = \int_3^5 g(y)(5 - y)g'(y)dy = \left((5 - y)\frac{g^2(y)}{2}\right)_3^5 + \int_3^5 \frac{g^2(y)}{2}dy$$

$$-2 \cdot \frac{g^2(3)}{2} + \frac{9}{2} = \frac{7}{2}$$

$$\text{Hence } 2 \int_1^4 x(x - f^{-1}(x))dx = 7$$

83. 7

$$\text{Sol. } S_k = \int_0^1 x^2(1-x)^k dx = \int_0^1 (1-x)^2 x^k dx \quad \left(\text{use } \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right)$$

$$\text{Hence } \sum_{k=1}^{\infty} \int_0^1 (1-x)^2 x^k dx = \int_0^1 (1-x)^2 \underbrace{\sum_{k=1}^{\infty} x^k}_{\text{infinite G.P.}} dx$$

$$= \int_0^1 (1-x)^2 \left(\frac{x}{1-x}\right) dx = \int_0^1 (x - x^2) dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \Rightarrow p + q = 7$$

84. 108

 Sol.  $\vec{l} = x\vec{m} + y\vec{n} + z\vec{k}$ 

$$\Rightarrow 1 = \frac{-1}{11}(x + y + z) \quad (\text{dot product with } \vec{l})$$

$$x + y + z = -11 \quad \dots(i)$$

$$\frac{-1}{11} = x - \frac{1}{11}(y + z) \quad (\text{dot product with } \vec{m})$$

$$1 + 11m = y + z \quad \dots (2)$$

$$\Rightarrow x = -1$$

$$\Rightarrow y(\vec{n} \cdot \vec{k}) + z = y + (\vec{n} \cdot \vec{k})z \quad (\text{dot product with } \vec{n} \text{ and } \vec{k})$$

$$\Rightarrow y = z \Rightarrow \vec{l} + \vec{m} = y(\vec{n} + \vec{k})$$

$$\Rightarrow y = -5; |\vec{l} + \vec{m}| = 5|\vec{n} + \vec{k}|$$

$$\Rightarrow 2 - \frac{2}{11} = 25(2 + 2\vec{n} \cdot \vec{k}) \Rightarrow 2 + 2\vec{n} \cdot \vec{k} = \frac{4}{55}$$

$$\Rightarrow \vec{n} \cdot \vec{k} = \frac{-53}{55}$$

85. 10

Sol. Use basic concepts.

86. 4

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{\ln((\cos x)^a)}{x^b} = \lim_{x \rightarrow 0} \frac{a \ln(\cos x)}{x^b}$$

Now applying L' Hospital rule

$$\lim_{x \rightarrow 0} \frac{-a \tan x}{bx^{b-1}} = \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \frac{-a}{b} \frac{1}{x^{b-2}} = \lim_{x \rightarrow 0} \frac{-a}{b} \frac{1}{x^{b-2}}$$

Now for limit to be finite

$$b - 2 = \{0, -1, -2, -3, -4, \dots\}$$

$$b = \{2, 1, 0, -1, -2, -3, \dots\}$$

 But b can only be  $b = \{2, 1\}$  as it is an outcome of a dice.

Now probability is

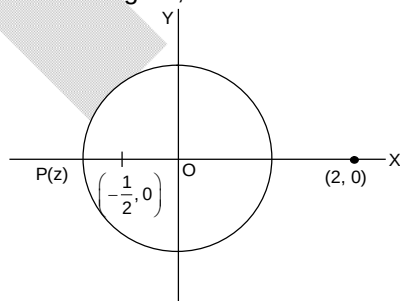
$$P = \frac{\text{No. of ways to select 'a'}}{\text{Total no. of ways to select 'a'}} \cdot \frac{\text{No. of ways to select 'b'}}{\text{Total no. of ways to select 'b'}}$$

$$P = \frac{6}{2} \cdot \frac{2}{3} = \frac{1}{3} \Rightarrow p + q = 1 + 3 = 4$$

87. 36

 Sol. Locus of z is  $x^2 + y^2 = 1$ 

From the figure, it is clear that maximum value of





$$2\left(\left|z_1 - 2\right| + \left|z_2 + \frac{1}{2}\right|\right) = 2 \times \frac{9}{2} = \lambda$$

$$\therefore 4\lambda = 36$$

88. 19

Sol.  $f(1) = 5, f(2) = 8, f'(1) = 3$  and  $f''(1) = 0$

$$f(x) = (x-1)^3(x-2) + 3x + 2$$

$$f(3) = 8 + 9 + 2 = 19$$

89. 3

$$\text{Sol. } P(A) = ({}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_{n-1})\left(\frac{1}{2}\right)^n = \frac{2^n - 2}{2^n}$$

$$P(B) = P(0 \text{ girl or girl}) = ({}^nC_0 + {}^nC_1)\left(\frac{1}{2}\right)^n = \frac{n+1}{2^n}$$

$$P(A \cap B) = P(\text{exactly one girl}) = {}^nC_1 \times \left(\frac{1}{2}\right)^n$$

$$\text{Now, } P(A \cap B) = P(A)P(B)$$

$$\frac{n}{2^n} = \frac{2^n - 2}{2^n} \left(\frac{n+1}{2^n}\right) \Rightarrow n = \frac{(2^n - 2)(n+1)}{2^n}$$

90. 0

Sol. Since point of minima is negative therefore point of maxima is also negative.

Hence, both roots  $f'(x)$  must be negative and distinct.

Sum of the roots  $< 0$  and  $D > 0$

Their intersection is  $\phi$ , hence no values of  $a$ .